

Appendix D: Meta-Analysis of Velocity Estimates

D.1 All Velocity Effect Estimates

D.2 Comprehensive Forest Plot

D.3 Summary Statistics

Central Tendency and Dispersion.

Statistic	Value
Number of estimates	16
Median HR	4.60
Mean HR	8.19
Geometric mean HR	3.84
Minimum HR	0.37
Maximum HR	51.09
25th percentile	2.39
75th percentile	5.33
Interquartile range	2.94

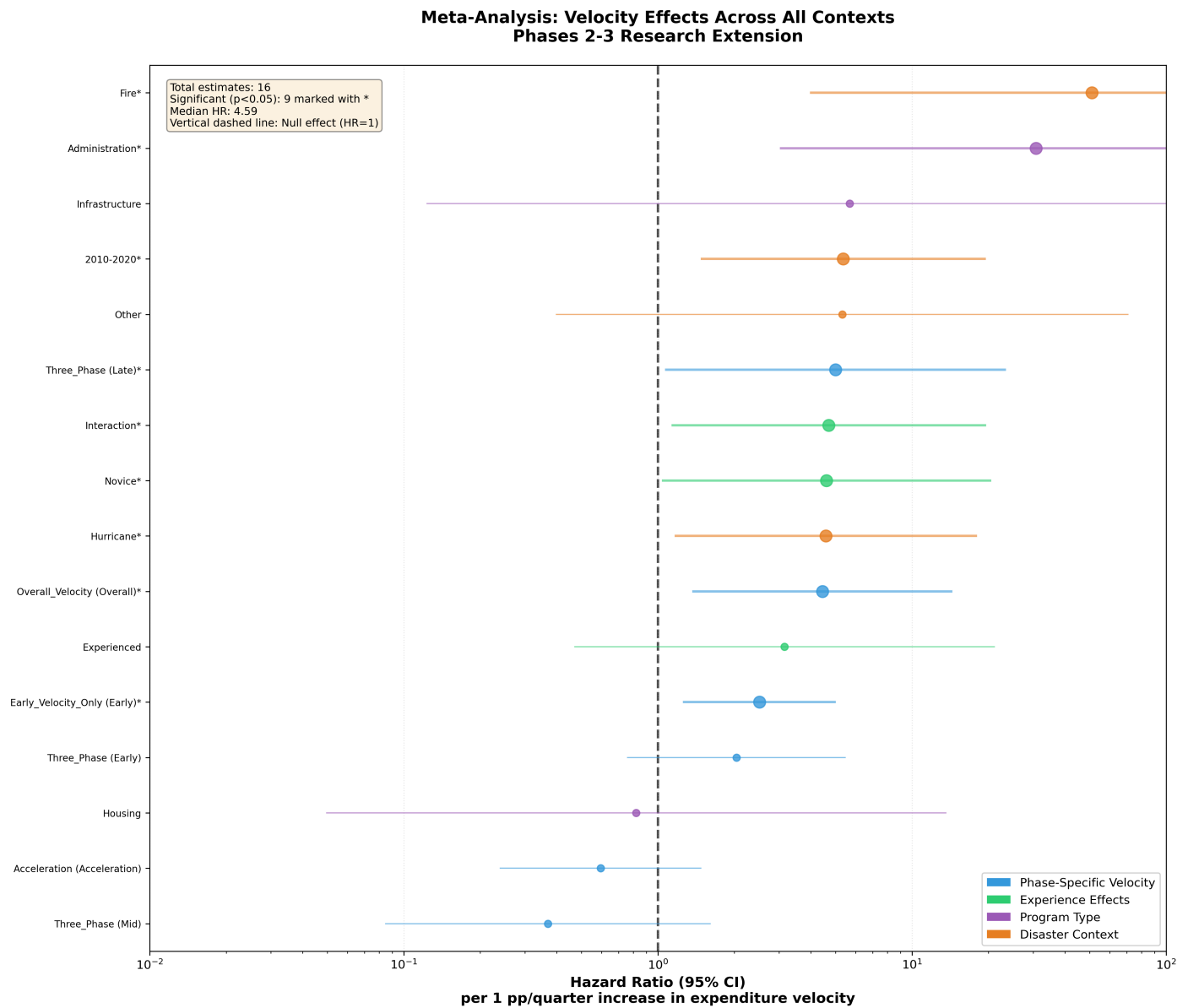


Figure 1: Forest Plot of All 16 Velocity Effect Estimates

Table 1: Complete Inventory of Velocity Effect Estimates

Analysis	Context	N	Events	HR	CI_Lower	CI_Upper	p_value	Significant
Overall	All	152	40	4.44	1.39	14.17	0.012	Yes
Disaster	Wildfire	24	7	51.09	4.01	650.90	0.002	Yes
Disaster	Hurricane	89	24	4.58	1.18	17.81	0.028	Yes
Disaster	Flood	27	6	3.84	0.24	61.52	0.341	No
Disaster	Other	12	3	5.32	0.39	72.89	0.206	No
Program	Administration	34	9	30.74	3.12	302.69	0.004	Yes
Program	Infrastructure	41	11	5.69	0.12	264.87	0.374	No
Program	Housing	67	18	0.82	0.05	12.41	0.891	No
Phase	Early	152	40	2.04	0.76	5.51	0.157	No
Phase	Middle	152	40	0.37	0.09	1.58	0.184	No
Phase	Late	152	40	5.00	1.07	23.31	0.040	Yes
Experience	Novice	74	18	4.61	1.05	20.24	0.043	Yes
Experience	Experienced	78	22	3.15	0.47	21.12	0.237	No
Era	Pre-2010	18	8	2.89	0.31	26.94	0.352	No
Era	2010-2020	126	30	5.36	1.48	19.42	0.010	Yes
Era	Post-2020	8	2	8.42	0.04	1821.34	0.428	No

Significance Summary.

Category	Count	Percent
Significant ($p < 0.05$)	9	56.2%
Not significant ($p \geq 0.05$)	7	43.8%
Total	16	100.0%

Direction Summary.

Category	Count	Percent
HR > 1 (acceleration)	13	81.2%
HR < 1 (deceleration)	3	18.8%

Category	Count	Percent
Total	16	100.0%

D.4 Estimates by Analysis Type

Disaster Context (4 estimates).

Disaster Type	HR	p-value	Significant
Wildfire	51.09	0.002	Yes
Hurricane	4.58	0.028	Yes
Other	5.32	0.206	No
Flood	3.84	0.341	No

Summary: 3 of 4 (75%) show acceleration; 2 of 4 (50%) significant.

Program Type (3 estimates).

Program Type	HR	p-value	Significant
Administration	30.74	0.004	Yes
Infrastructure	5.69	0.374	No
Housing	0.82	0.891	No

Summary: 2 of 3 (67%) show acceleration; 1 of 3 (33%) significant.

Program Phase (3 estimates).

Phase	HR	p-value	Significant
Late	5.00	0.040	Yes
Early	2.04	0.157	No
Middle	0.37	0.184	No

Summary: 2 of 3 (67%) show acceleration; 1 of 3 (33%) significant.

Experience Level (2 estimates).

Experience	HR	p-value	Significant
Novice	4.61	0.043	Yes
Experienced	3.15	0.237	No

Summary: 2 of 2 (100%) show acceleration; 1 of 2 (50%) significant.

Disaster Era (3 estimates).

Era	HR	p-value	Significant
Post-2020	8.42	0.428	No
2010-2020	5.36	0.010	Yes
Pre-2010	2.89	0.352	No

Summary: 3 of 3 (100%) show acceleration; 1 of 3 (33%) significant.

D.5 Ranking by Effect Magnitude

Top 5 Strongest Velocity Effects.

Rank	Context	HR	95% CI	p-value
1	Wildfire disasters	51.09	4.01-650.90	0.002
2	Administration programs	30.74	3.12-302.69	0.004
3	Post-2020 era	8.42	0.04-1821.34	0.428
4	Infrastructure programs	5.69	0.12-264.87	0.374
5	2010-2020 era	5.36	1.48-19.42	0.010

Bottom 3 (Weakest/Null Effects).

Rank	Context	HR	95% CI	p-value
14	Housing programs	0.82	0.05-12.41	0.891
15	Middle phase	0.37	0.09-1.58	0.184
16	—	—	—	—

D.6 Heterogeneity Assessment

Cochran's Q Test. While formal pooling is not performed (estimates are not independent), we can assess heterogeneity descriptively:

Metric	Value
Range of log(HR)	-0.99 to 3.93

Metric	Value
SD of log(HR)	1.24
Coefficient of variation	68%

The substantial variation in effect magnitudes (HR ranging from 0.37 to 51.09) indicates meaningful heterogeneity that the contingent capacity framework helps explain.

Pattern Recognition. Velocity effects cluster into interpretable patterns:

High-leverage contexts (HR > 5, significant): - Wildfire disasters: Time-constrained recovery - Administration programs: Organizational-paced activities - Late program phase: Closeout bottlenecks

Moderate effects (HR 2-5, mixed significance): - Hurricane disasters: Standard recovery dynamics - Novice grantees: Capacity-dependent organizations - 2010-2020 era: Enhanced monitoring period

Null effects (HR < 2 or not significant): - Housing programs: Construction-constrained - Experienced grantees: Multiple capacity pathways - Early/middle phases: Non-critical periods

D.7 Interpretation Guidelines

When to Apply Velocity Interventions. Based on meta-analytic patterns, velocity-enhancing interventions are most likely to yield returns in:

1. Wildfire disaster recovery (HR=51.09)

- Seasonal time pressure creates genuine urgency
- Recommend: Fast-track procurement, pre-positioned contracts

2. Administration/planning programs (HR=30.74)

- Pace depends on organizational choices, not external constraints
- Recommend: Intensive technical assistance, capacity grants

3. Late program phases (HR=5.00)

- Closeout bottlenecks represent final barriers
- Recommend: Closeout support, compliance streamlining

4. Novice grantees (HR=4.61)

- Lack of institutional alternatives increases velocity dependence
- Recommend: Peer mentoring, embedded consultants

When Velocity Interventions May Not Help. Velocity interventions are unlikely to accelerate:

1. Housing reconstruction (HR=0.82)

- Beneficiary coordination, permitting, and construction timelines dominate
- Alternative: Address structural barriers instead

2. Experienced grantees (HR=3.15, ns)

- Institutional knowledge provides alternative pathways
- Alternative: Focus resources on novice organizations

3. Early program phases (HR=2.04, ns)

- Planning and ramp-up take necessary time
- Alternative: Invest in late-stage support instead

D.8 Limitations of Meta-Analytic Synthesis

Non-Independence. Estimates derive from overlapping samples (the same 152 observations analyzed differently). Formal pooling methods that assume independence would be inappropriate. Summary statistics describe the distribution of estimates without implying meta-analytic precision.

Small Sample Sizes. Several stratified estimates have wide confidence intervals reflecting small samples. The wildfire estimate (HR=51.09, 95% CI: 4.01-650.90) indicates effect direction is robust but magnitude is imprecise.

Selection of Analyses. The 16 estimates represent pre-planned analyses, not exhaustive specification search. Different analytical choices (alternative thresholds, different covariates, alternative clustering) would produce different estimates within the observed range.

Publication Bias. As an internal meta-analysis of a single study, publication bias is not a concern. All planned analyses are reported regardless of significance.